BrickMore

Authors and affiliations to be added

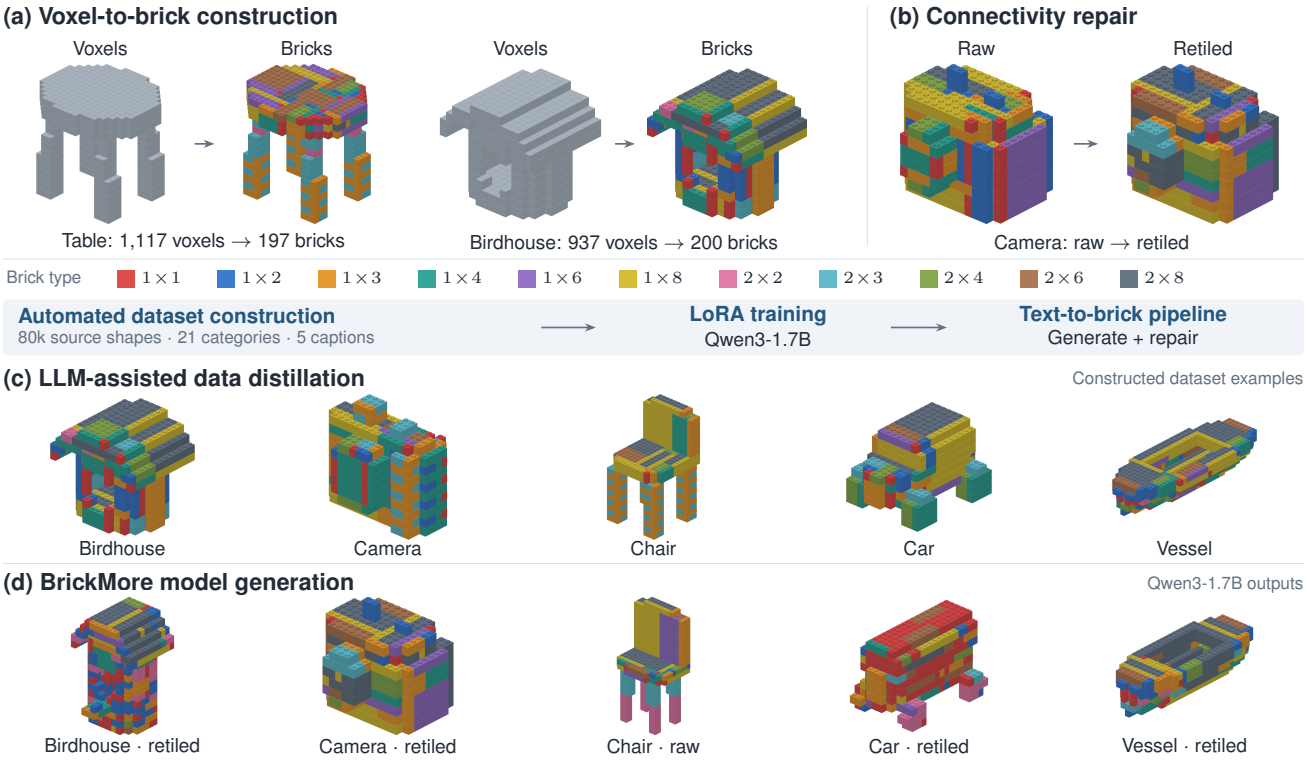


Figure 1. **BrickMore overview.** (a) Source voxels are converted into rectangular bricks. (b) Raw predictions are retiled to repair connectivity. (c) Examples from agentic batch generation of voxel data. (d) Outputs of the fine-tuned Qwen3-1.7B pipeline. Colors consistently encode brick type, independent of orientation; they are visualization labels, not predicted appearance. Voxels are shown in gray.

Abstract

Generating brick structures from text requires both alignment with natural language and valid geometric connections between discrete pieces. We present BrickMore, a text-to-brick framework that combines large-language-model-assisted data construction, language-model fine-tuning, and post-generation geometric repair. Using GPT-6.1 Sol, we develop an agentic batch-generation pipeline in which the model writes parametric generator programs that then produce three-dimensional voxel models in batches. The resulting shapes are converted into brick layouts through same-color tiling, connectivity analysis, and local bridging and retiling. The source dataset contains approximately 80,000 voxel models across 21 object categories on a 48^3 grid. Each model has five English descriptions at different levels of detail, for approximately 400,000 descriptions in total. In the latest integrated version, nearly 80,000 models pass the specified geometric checks. We fine-tune Qwen3-1.7B using low-rank adaptation (LoRA) on English descriptions and brick-placement sequences, enabling it to predict brick dimensions and positions from text.

After generation, the pipeline forms a voxel union from the in-bounds portions of supported brick placements, uses mixed-integer linear programming (MILP) to seek an exact cover, and checks for a single stud-connected component. We evaluate the resulting structures against outputs from BrickGPT and BrickNet on 114 shared prompts. On the evaluated outputs, BrickMore achieves the highest mean scores across three text–image alignment metrics. It also obtains a mean invalid-placement ratio of 0.00%, matching BrickGPT under the same metric. These comparisons use a shared rendering protocol.

1. Introduction

Text-to-three-dimensional generation enables users to describe and create objects using natural language. Generating brick structures adds the requirement that a target shape be represented as discrete pieces and their spatial placements. In addition to matching the text, the resulting struc-

ture must account for overlaps and connections between bricks: a layout may resemble the described object while containing intersecting bricks or disconnected parts, and thus fail to meet the specified geometric requirements.

Prior work has explored text-driven brick generation through different representations and generation constraints. BrickGPT trains an autoregressive language model on paired text and brick structures and applies validity checks and physics-aware rollback during inference (Pun et al. 2025). BrickNet uses a graph-based representation to generate assembly sequences involving a broader range of parts and connection types (Kulits and Schmid 2026). These works demonstrate the importance of structural representations and generation constraints for brick modeling.

For voxel-based shapes, spatial adjacency between voxels does not necessarily imply a valid connection between bricks. A voxel shape can form a connected volume even when the bricks covering it only touch side by side without stud connections. This distinction affects both data construction and generation: converting voxel shapes into training targets requires attention to coverage and brick connectivity, while predicted layouts may also contain overlaps or disconnected components that require further processing.

To study this problem, we present BrickMore. Using GPT-6.1 Sol, we develop an agentic batch-generation pipeline: the model writes parametric generator programs, which then produce voxel models in batches, and we convert the resulting models into brick layouts. The source dataset contains 80,000 models across 21 object categories, represented on a 48^3 grid, with five levels of English description for each model. We tile occupied voxels with same-color rectangular bricks and combine connection analysis with local bridging and retiling to address ungrounded components. Some derived targets involve limited changes to the source voxels; these changes and the original data are recorded separately. In the latest integrated version, 79,301 models pass the specified geometric checks.

We fine-tune Qwen3-1.7B with LoRA on English descriptions and brick-placement sequences. Given a text prompt, the model predicts brick dimensions and positions; the current output does not include color. To address potential overlaps and connectivity defects, we form a target union from the in-bounds voxels of supported brick placements, use MILP to seek an exact cover, and check for a single stud-connected component. For some cases in which no feasible layout is found within the solver's time budget, the system regenerates a sequence using a different language-model sampling seed. The connectivity check is geometric; physical stability and the feasibility of an assembly order require separate evaluation.

To evaluate the resulting structures, we compare the provided outputs of BrickMore, BrickGPT, and BrickNet on 114 shared prompts across 21 object categories and five description levels. We measure text–image alignment using VQAScore, PE, and SigLIP2, and evaluate the proportion of invalid brick placements using P_{inv}. Under a common rendering protocol, BrickMore achieves the highest mean scores on all three text–image metrics. BrickMore and BrickGPT both obtain a mean P_{inv} of 0.00% when evaluated in file order. These results describe the provided outputs under the specified evaluation protocol and do not independently establish physical stability or assembly feasibility.

Our contributions are as follows:

1. We develop an automated 48^3 voxel data-generation pipeline by agentic batch generation with GPT-6.1 Sol, which writes parametric generator programs that produce 80,000 source models across 21 categories, each with five levels of English description. In the latest integrated version, 79,301 models pass the specified geometric checks.
2. We develop a voxel-to-brick conversion pipeline that combines same-color tiling, connection analysis, and local bridging and retiling, while recording limited changes to source voxels in derived targets.
3. We fine-tune Qwen3-1.7B for text-to-brick generation and combine it with a post-generation pipeline based on voxel unions, exact-cover MILP, and connectivity checks.

2. Related Work

Text-to-brick generation. BrickGPT represents a design as an autoregressively generated sequence of brick placements and uses brick-level rejection sampling together with physics-aware rollback during inference (Pun et al., 2025) [1]. BrickNet instead represents part connectivity through a graph-backed parametrization and trains autoregressive models on its assembly sequences (Kulits and Schmid, 2026) [2]. BrickMore also generates placement sequences from text, then examines geometric reconstruction and connectivity in a separate post-generation stage.

Stability analysis. StableLego studies the structural stability of block-stacking assemblies using an optimization formulation based on force balance (Liu et al., 2024) [3]. Such analyses depend on their mechanical assumptions. In BrickMore, geometric connectivity and the contact-based screening described in the dataset materials address different properties; neither alone establishes that every generated model can be physically assembled and remains stable.

3. Dataset Construction

3.1. Parametric Voxel Generation

The source dataset contains 80,000 voxel models on a 48^3 grid across 21 object categories. Each model is paired with five English descriptions at different levels of detail.

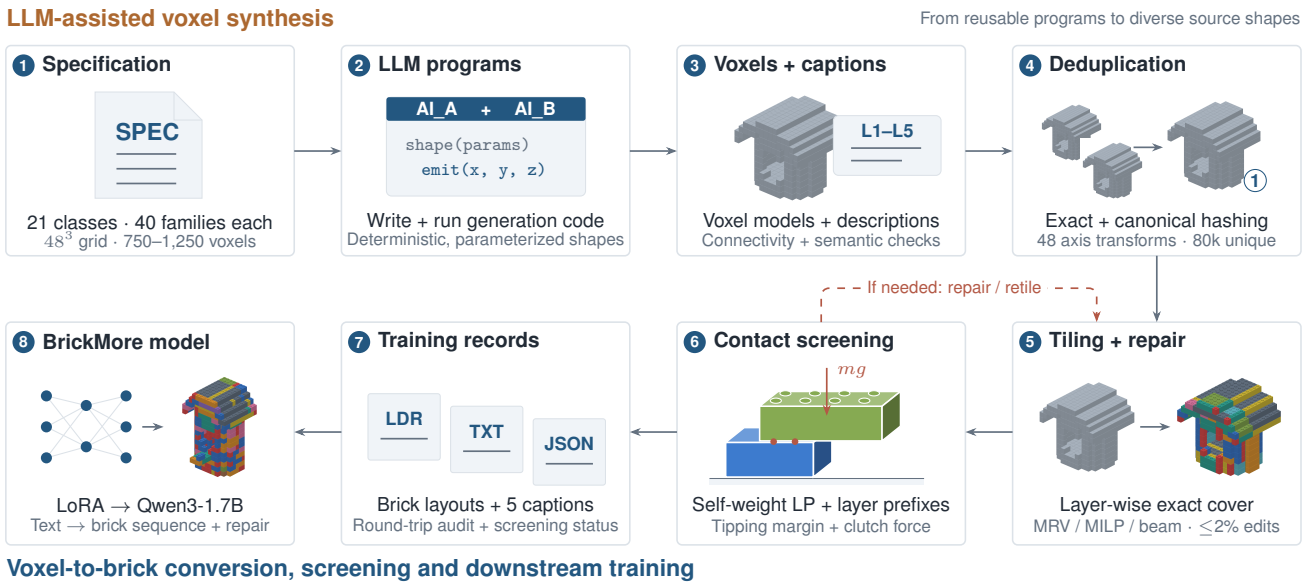


Figure 2. **Visual data-construction workflow.** Parametric programs written by GPT-6.1 Sol through agentic batch generation produce voxel shapes and descriptions, followed by deduplication, voxel-to-brick tiling, repair, contact-based screening, and audited export for downstream fine-tuning. The dashed arrow denotes repair or retiling when needed. This diagram summarizes the historical construction workflow; screening is distinct from a guarantee of physical stability. Brick colors follow the type legend in Figure 1.

The models are produced by parametric generator programs written through an agentic batch-generation process with GPT-6.1 Sol; the programs, rather than the language model itself, emit the voxel coordinates in batches. Model generation records and version information should be retained to support reproduction.

3.2. Voxel-to-Brick Conversion

The conversion pipeline tiles occupied voxels with same-color rectangular bricks and analyzes their connections. Local bridging and retiling are used to address ungrounded components, with changes to source voxels recorded separately. The historical relaxed and retile dataset variants may contain multiple components that are individually grounded; this criterion differs from the single-component check used in the post-generation pipeline.

3.3. Geometric Checks and Physical Screening

Geometric checks assess properties of the constructed layout under the relevant release rules. The reference materials also describe a contact-based gravity screening procedure under specified assumptions. Its outputs are screening results rather than proof of full physical stability; assumptions, coverage, and status counts for the exact dataset version require verification before being reported as final results.

3.4. Dataset Output and Audit

The historical full retile report records 79,298 geometrically passing models. Replacing the original 1,905 batch-B piano slots, of which 1,902 passed, with the updated release containing 1,905 passing models yields an integrated accounting of 79,301 passing models. This number is computed from the two release reports; it is not a new full-dataset verification run. The original A/B download procedure does not automatically perform this replacement.

4. Dataset Statistics and Analysis

The historical and integrated figures represent different version boundaries. The rules used by each release and the source of the integrated figure should accompany any comparison. The actual training manifest remains to be checked before identifying the version used to train Qwen3-1.7B.

4.1. Physical Screening Results

Physical screening results should be reported separately from geometric pass rates, with the relevant assumptions, assessed sample count, and unresolved cases identified. Historical counts from different releases should not be combined without a version-matched audit.

5. BrickMore-48 Text-to-Brick Model

We fine-tune Qwen3-1.7B using LoRA on English descriptions and brick-placement sequences. Given a text prompt,



Figure 3. Voxel model samples from the reference document, spanning 21 object categories.

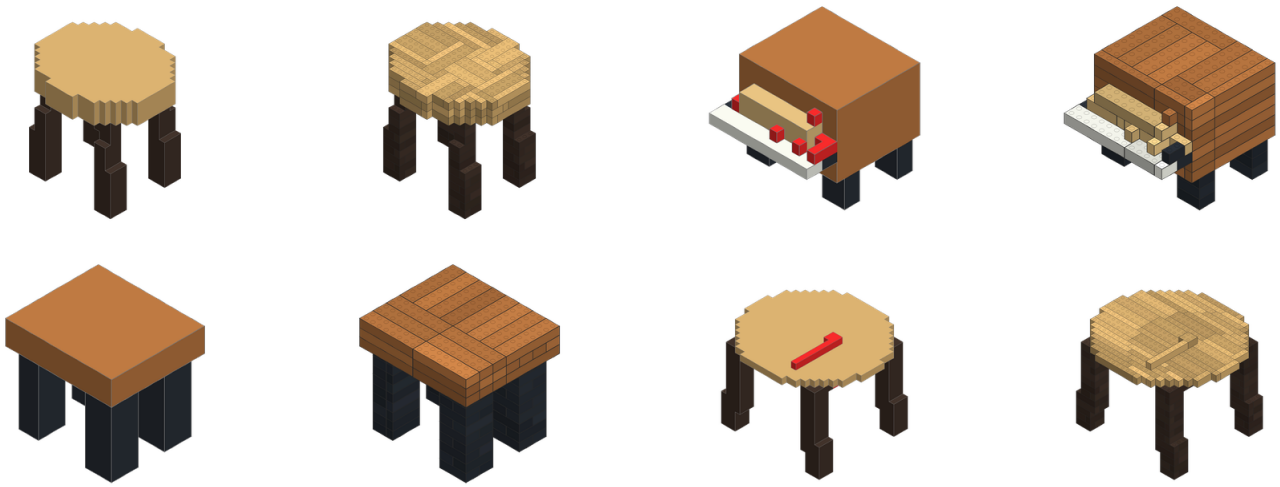


Figure 4. Historical examples of voxel targets and brick layouts after local repair.

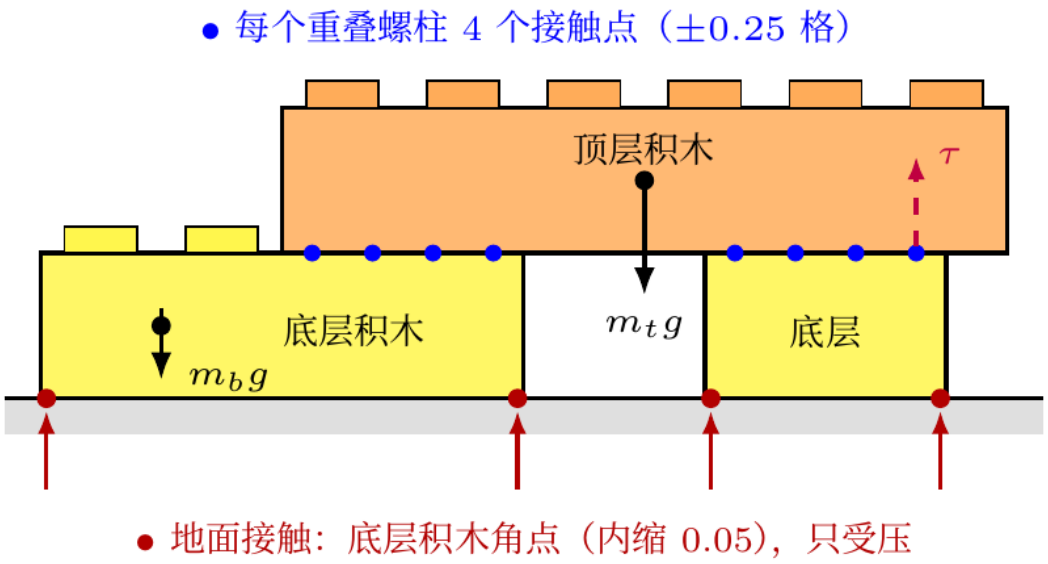


Figure 5. Illustration of the historical contact-based gravity screening model; source labels are retained.

the model predicts dimensions and positions of bricks. The current output does not include color. Training hyperparameters, data splits, training resources, and the precise training dataset version remain to be documented from the original

Item	Value
Source models	80,000
Categories	21
Grid	48^3
English descriptions	Approximately 400,000
Historical retile geometry passes	79,298 / 80,000
Integrated accounting	79,301 / 80,000

Table 1. Dataset statistics and version boundaries.

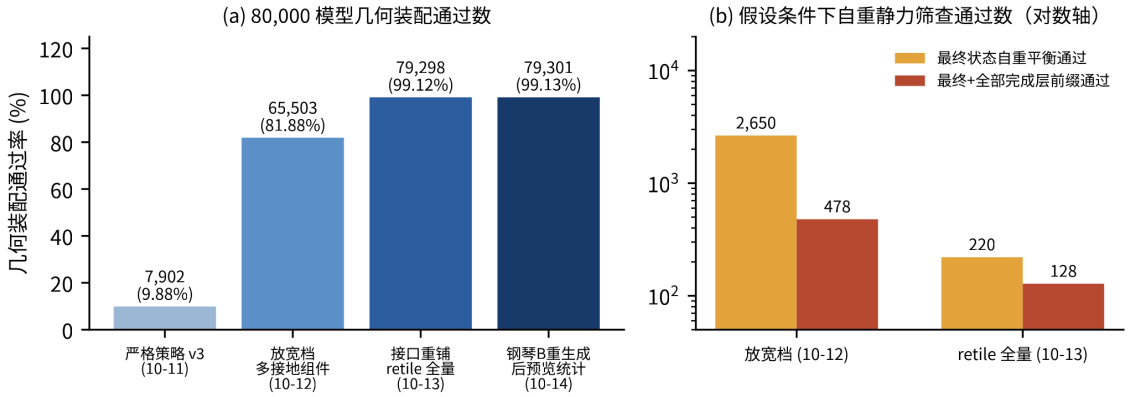


Figure 6. Historical comparison of geometric conversion and screening outcomes. Version and numerical labels require verification.

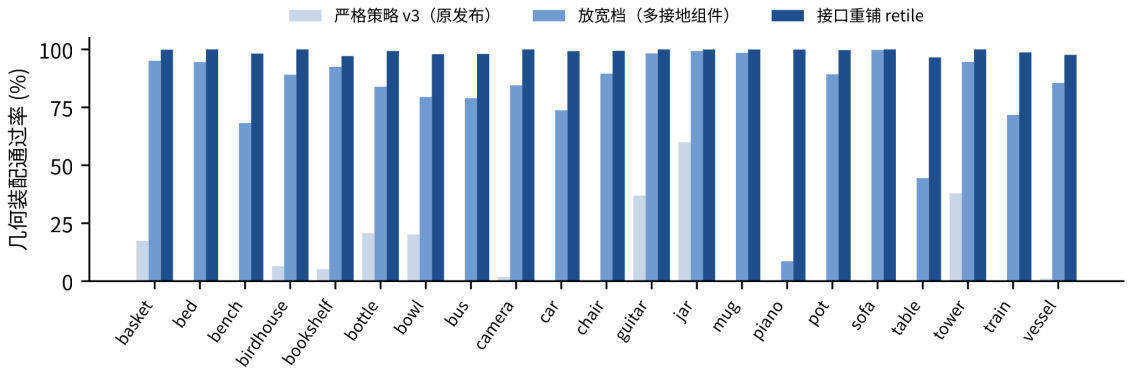


Figure 7. Historical category-level geometric pass rates. Version and numerical labels require verification.

training records.

Inference follows a seed-escalation and post-generation repair procedure, summarized in Algorithm 1. The model samples a placement sequence with thinking disabled, temperature 0.7, top-p 0.9, and a limit of 8192 new tokens, using seeds 42, 104, and 2026 in that order. A candidate is accepted when generation completes and the geometry validator reports no errors. If no such candidate exists, the pipeline keeps the completed draft with the fewest validator error records, denoted $\mathrm{Err}(\cdot)$, which counts line-format, library-membership, bounds, and overlap errors. It then forms a voxel union from the in-bounds cells of supported library placements, solves a shape-preserving exact cover with a single stud-connected component using mixed-integer linear programming (MILP) under a 12s–60s–180s time ladder, and runs an independent audit. If no feasible cover is found within this budget, the raw draft is retained and regeneration with a new seed may be queued.

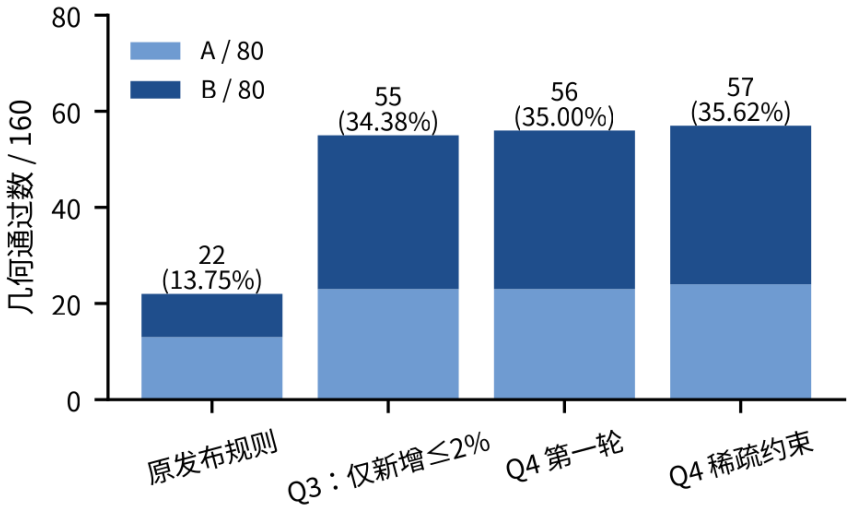


Figure 8. Historical results on a 160-model sample. Version and numerical labels require verification.

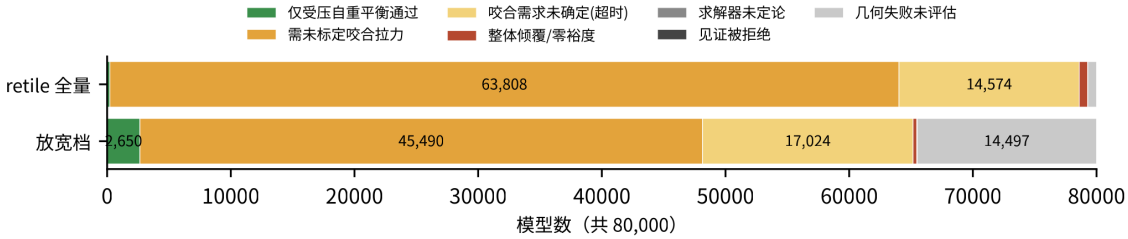


Figure 9. Historical distributions of physical screening statuses. Counts refer to the specific releases shown.

6. Comparative Evaluation

6.1. Setup and Metrics

We compare the provided outputs of BrickMore, BrickGPT, and BrickNet using 114 shared prompts covering 21 object categories and five description levels. The evaluation uses a common rendering protocol. Text–image alignment is assessed with VQAScore, PE, and SigLIP2; P_{inv} measures the proportion of invalid placements under the evaluation procedure.

The report specifies eight rendered views per model and aggregates each text–image metric using the maximum across views before averaging over models. P_{inv} is averaged per model and evaluated in exported file order. The supplied outputs retain differences in candidate selection, filtering, and post-processing; the comparison therefore characterizes these evaluated outputs rather than an isolated model-only comparison.

6.2. Main Results

On these outputs, BrickMore has the highest mean on each of the three text–image metrics. BrickMore and BrickGPT both have a mean P_{inv} of 0.00%; this measure does not distinguish them in the main comparison. Against BrickGPT, BrickMore scores higher on 91, 95, and 91 prompts for VQAScore, PE, and SigLIP2, respectively. The supplied analysis does not report a significance test.

The mean numbers of parts are 413.0 for BrickMore, 152.3 for BrickGPT, and 23.3 for BrickNet. These differences describe the evaluated structures and should be considered when interpreting the scores; part count alone does not measure generation quality.

6.3. Raw and Stitched Outputs

The supplied evaluation contains 110 matched raw and stitched BrickMore outputs. This paired subset is separate from the three-system main table. We match each pair by prompt, model identifier, and caption, and compare the mean scores before and after retiling.

System	$P_{\text{inv}} \downarrow$	VQAScore ↑	PE ↑	SigLIP2 ↑
BrickMore	0.00%	0.2176	0.1856	0.1231
BrickGPT	0.00%	0.1401	0.1457	0.0297
BrickNet	21.99%	0.0624	0.1302	0.0100

Table 2. Main comparative results on 114 matched prompts.

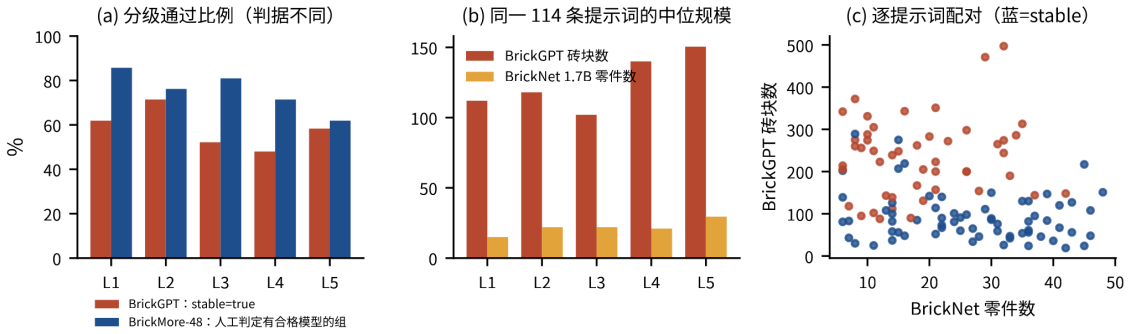


Figure 10. Historical baseline comparison with differing acceptance criteria; not the current matched main comparison.

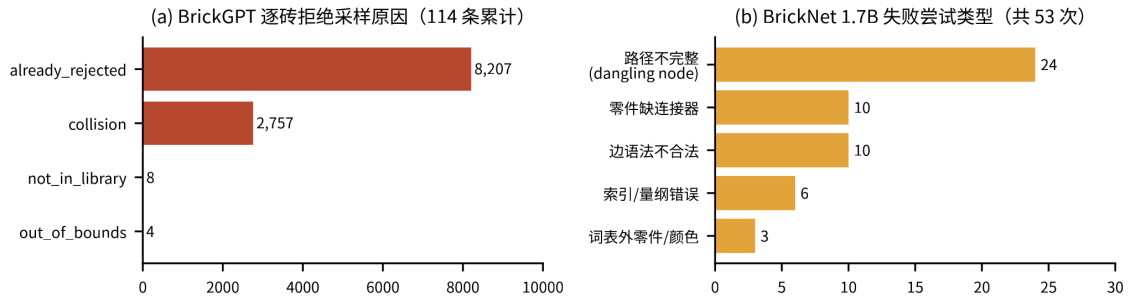


Figure 11. Historical failure categories in the supplied reference; sample definitions require verification.

Metric	Raw	Stitched
$P_{\text{inv}} \downarrow$	0.1897%	0.0000%
VQAScore ↑	0.2065	0.2071
PE ↑	0.1858	0.1859
SigLIP2 ↑	0.1379	0.1261

Table 3. Paired raw and retiled results on 110 matched cases.

The mean invalid-placement ratio decreases to zero after retiling. VQAScore and PE change only slightly, while the mean SigLIP2 score decreases. These are descriptive results for the matched subset and do not establish a general causal effect.

6.4. Evaluation Boundaries

PE truncates 25 of the 114 main-comparison prompts, and SigLIP2 truncates 29; the affected prompts are the same across the three systems for each metric. The evaluation uses a common yellow rendering and therefore does not assess original colors. A zero P_{inv} does not establish stud connectivity, physical stability, or assembly feasibility.

7. Limitations and Discussion

The integrated dataset count is derived from versioned release reports, and the version actually used for training needs confirmation. Dataset conversion may permit multiple grounded components, while the generation-stage

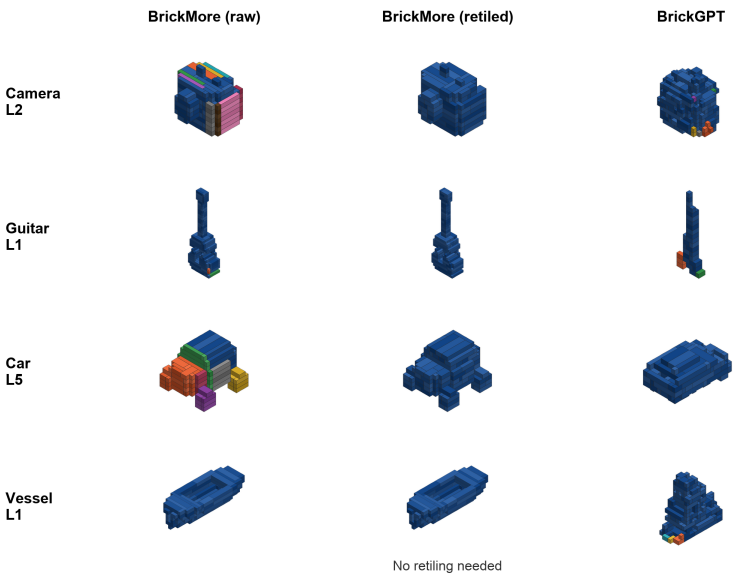


Figure 12. Qualitative historical examples from the supplied reference. Column labels distinguish raw BrickMore, retiled BrickMore, and BrickGPT outputs.

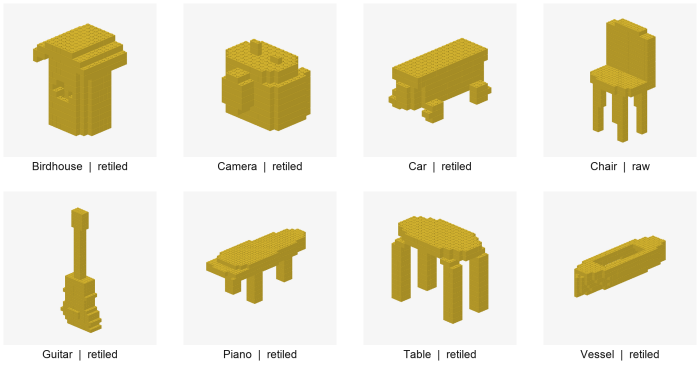


Figure 13. Selected BrickMore outputs from the 114-prompt evaluation. One chair is raw; the other outputs are retiled. Results use uniform evaluation rendering and are not photographs of physical assemblies.

single-component check has a different scope. Physical screening is based on stated assumptions and does not replace calibrated clutch-strength measurement or physical assembly tests.

The comparative evaluation covers 114 prompts and the provided outputs. The systems differ in their candidate selection and post-processing, and some text is truncated by evaluation models. These conditions limit how broadly the reported scores can be interpreted. Further paired uncertainty analysis can be conducted on the existing per-prompt records.

8. Conclusion

We present BrickMore, a text-to-brick framework combining LLM-assisted data construction, language-model fine-tuning, and post-generation geometric repair. The dataset workflow converts programmatically generated voxel shapes into brick layouts, while the generation pipeline predicts placements from text and checks reconstructed layouts. On 114 shared prompts, the evaluated BrickMore outputs achieve the highest means on three text–image metrics, while matching BrickGPT on mean invalid-placement ratio. The results should be interpreted under the stated evaluation and physical-screening boundaries.

Algorithm 1 BrickMore-48 inference and retiling

Require: text prompt c; model θ (Qwen3-1.7B fine-tuned on BrickMore-48); brick library $\mathcal{L}$ (20 types); seeds $S = \{42, 104, 2026\}$; MILP time tiers $T = \{12, 60, 180\}$ seconds

Ensure: a pair $(B_{\text{out}}, \textit{status})$ with $\textit{status} \in \{\text{SUCCESS}, \text{DRAFT}, \text{FAILED}\}$

$B \leftarrow \emptyset$; $\textit{attempts} \leftarrow \emptyset$
for $s \in S$ **do**
  $\textit{ctx} \leftarrow \text{Chat}(c,\ \textit{thinking} = \text{false})$
  $T_s \leftarrow \theta.\text{Generate}(\textit{ctx},\ \text{seed} = s)$ ▷ $t = 0.7$, top-$p = 0.9$, max-new $= 8192$
  $(B_s,\ \textit{rep}) \leftarrow \text{Parse}(T_s)$ ▷ lines $h \times w$ (x, y, z)
  $\textit{attempts} \leftarrow \textit{attempts} \cup \{(s, B_s, \textit{rep})\}$
  if $\textit{rep}.\text{complete} \wedge \textit{rep}.\text{geometry_valid}$ **then**
    $B \leftarrow B_s$
    break
  end if
end for
if $B = \emptyset$ **then**
  if $\{a \in \textit{attempts} : a.\textit{rep}.\text{complete}\} = \emptyset$ **then**
    return $(\emptyset,\ \text{FAILED})$
  end if
  $B \leftarrow \arg\min_{a:\ a.\textit{rep}.\text{complete}} \text{Err}(a)$ ▷ fewest validator error records
end if
$C \leftarrow \text{UnionVoxels}(B;\ \mathcal{L},\ 48^3)$ ▷ whitelist and clip; overlaps count as one voxel
$B^* \leftarrow \emptyset$
for $\tau \in T$ **do**
  $B^* \leftarrow \text{MILP-SingleStudCover}(C,\ \tau)$
  if $B^* \neq \emptyset$ **then**
    break
  end if
end for
if $B^* \neq \emptyset$ **then**
  $\text{Export}(B,\ B^*)$ ▷ `raw.txt`, `stitched.txt`, and `*.ldr`
  $\text{Audit}(B^*,\ C)$ ▷ one stud component and $\text{voxels}(B^*) = C$
  return $(B^*,\ \text{SUCCESS})$
else
  $\text{QueueReseed}(c)$; keep B ▷ unsolved: retain the raw draft
  return $(B,\ \text{DRAFT})$
end if

$\text{Err}(\cdot)$ counts records from the geometry validator (line format, library membership, bounds, and overlaps). A completed sequence is one that ends with an end-of-sequence token. The default ladder is $T = \{12, 60, 180\}$. A later recovery path may extend the solver budget to 600 s and sample additional seeds outside S; those settings are not part of the default loop above.

References

[1] A. Pun, K. Deng, R. Liu, D. Ramanan, C. Liu, and J.-Y. Zhu. Generating physically stable and buildable brick structures from text. In *ICCV*, 2025. https://arxiv.org/abs/2505.05469.

[2] P. Kulits and C. Schmid. BrickNet: Graph-backed generative brick assembly. In *CVPR*, 2026. https://arxiv.org/abs/2604.22984.

[3] R. Liu, K. Deng, Z. Wang, and C. Liu. StableLego: Stability analysis of block stacking assembly. *IEEE Robotics and Automation Letters*, 9(11):9383–9390, 2024. https://doi.org/10.1109/LRA.2024.3414281.

A. Historical Tables from the Supplied Reference

The following transcribed tables preserve the historical reference material. They use different versions or evaluation criteria and are not substitutes for the current main comparison. Values, physical-screening coverage, and missing entries require verification. Dashes indicate missing entries in the supplied source.

Item	Historical value
Voxel models	80,000 (40,000 per batch)
Categories / structural families	21 / 40 per category
Grid / voxel count	48^3 / 750–1,250
Prompt records	400,000 (five bilingual records per model)
Uniqueness	80,000 unique; 0 A/B duplicates
Overlap with historical hashes	0 / 55,255
Retile geometry passes	79,298 (99.12%)
Bricks in full preview	12,718,389
Final gravity equilibrium passes	220
Final and layer-prefix passes	128

Table 4. Historical dataset statistics transcribed from the supplied reference. The geometry count refers to the old retile release, not the later integrated count. Verify the preview and physical-screening totals against version-matched reports.

Version	Connectivity	Voxel changes	Passes	Rate	Final	Final + prefixes
Strict v3	Single; support $\geq$10%, overhang $\leq$2	None	7,902	9.88%	–	–
Relaxed	Each component grounded	+428,278 / -38,230	65,503	81.88%	2,650	478
Retile	Each component grounded	+228,582 / -125	79,298	99.12%	220	128
Uniform 160-model sample (seed 38177); not full-dataset results						
Baseline	Single stud-connected component	None	22	13.75%	–	–
Q3	Single stud-connected component	Added $\leq$2%	55	34.38%	8	2
Q4 (sparse)	Single stud-connected component	Added + removed $\leq$2%	57	35.63%	8	2

Table 5. Historical conversion-strategy table from the reference. Full-dataset and 160-model sample rows use different scopes; connectivity rules also differ. Final and prefix columns report screening outcomes under the stated historical assumptions.

Item	BrickGPT	BrickNet 1.7B	BrickMore-48
Representation	20^3 bricks	Assembly graph	48^3 bricks
Criterion	stable	Parseable	–
Passes	66/114	114/114	–
Median parts	126.5	22	–
Range	19–497	6–48	–
Mean time (s)	189.6	–	–

Table 6. Historical baseline table in the source reference. These criteria differ, and missing BrickMore entries were marked TODO. This is not the current matched four-metric main table.

Level	n	BrickGPT stable	BrickGPT median bricks	BrickNet median parts	BrickMore-48
L1	21	13	112	15	–
L2	21	15	118	22	–
L3	23	12	102	22	–
L4	25	12	140	21	–
L5	24	14	150.5	29.5	–

Table 7. Historical level-wise baseline table transcribed from the reference. The original BrickMore column was incomplete, and the other columns use different criteria.

Method and source	P_{inv}	VQA	PE	SigLIP2
BrickGPT (literature, as transcribed)	0.063	0.050	0.157	0.052
BrickNet 1.7B (literature, as transcribed)	0.260	0.593	0.282	0.631
BrickGPT (114 prompts, historical placeholder)	–	–	–	–
BrickNet (114 prompts, historical placeholder)	–	–	–	–
BrickMore-48 (114 prompts, historical placeholder)	–	–	–	–

Table 8. Original reference table with literature values and historical placeholders. Literature values use another evaluation set and must not be directly compared with the present 114-prompt results; the precise P_{inv} definition and direction require checking against the original publication.